GLOBAL HR FORUM

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Vocational Education and Training in Times of AI
Chances and Challenges

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Figures from Germany - Trainees meet AI in two Ways

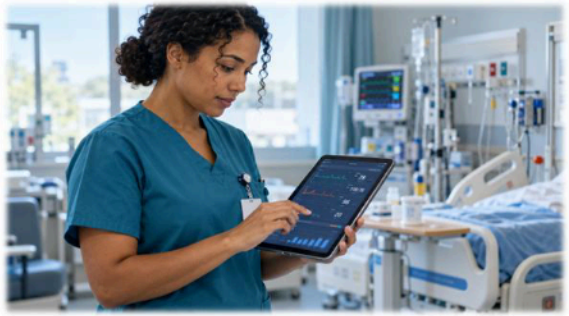
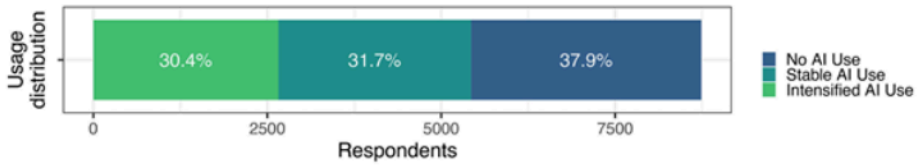




Figure 2: Distribution of AI Use by Usage Type



Note: Percentage distribution of respondents by type of artificial intelligence (AI) use at the workplace (No AI Use, Stable AI Use, Intensified AI Use).

Arntz, M., Baum, M., Brüll, E., Dorau, R., Hartwig, M., Lehmer, F., Matthes, B., M eyer, S.-C., Schlenker, O., Tisch, A., & Wischniewski, S. (2025). Digital transform ation and the changing world of work (DiWaBe 2.0): A data source for research on artificial intelligence and other technologies in the workplace. Bundesanstalt für Ar beitsschutz und Arbeitsmedizin, p.16.



Risius, P., & Hornickel, S. (2026). KI in der Ausbildung bereits häufig im Einsatz (IW-Kurzbericht Nr. 47/ 2026). Institut der deutschen Wirtschaft, p. 2.

Chances and Challenges

Usage of AI for work and during training leads to chances and challenges concerning VET





Chances: Support for Learning, Support for Teaching, Support for Research and Development

Challenges: Data-Security, Assessment, Competences (need and loss)

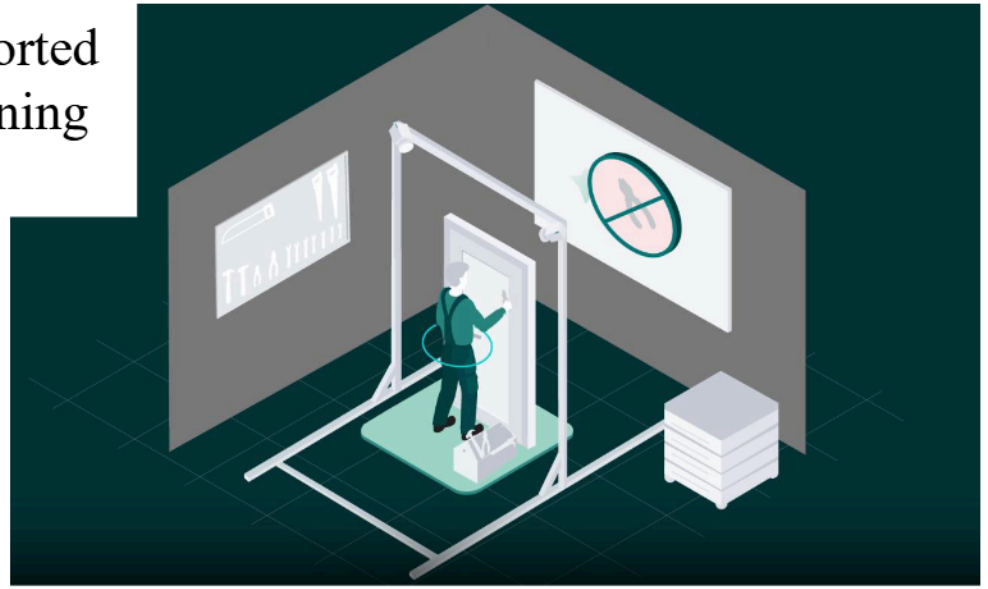
Chances



1. Support for Learning & Teaching: AI-supported tailored software for learning, possibilities to practise or create with LLM, possibility to create exercises or illustrative explanations

2. Support for Research and Development: analysing skill requirements, preparing research data, accessibility and summary of large texts

Example: AI-supported inter-company training for metalworkers



Source: project website: Home - KIgÜBA – KI-gestützte überbetriebliche Ausbildung

Challenges: Data-Security



Some applications of AI in education are classified as high-risk areas and therefore require special measures

European Commission

AI Act

(entered into force 2024)

Challenges: Assessment

Challenge: **Assessment**

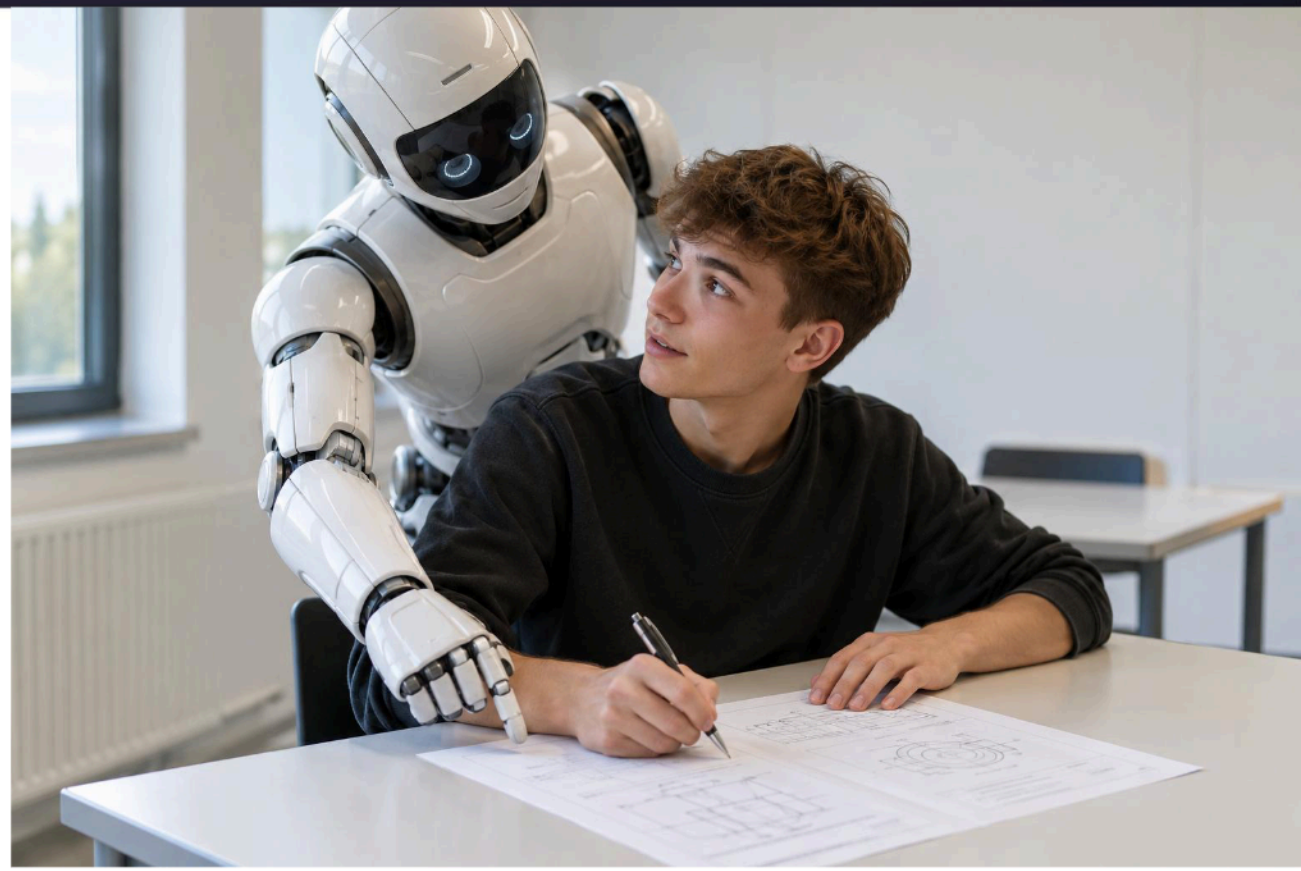
Vocational education and training under the BBiG

develops

Occupational competence

is assessed and demonstrated through

Final assessment / examination



Challenges: Competences



"Providers and deployers of AI systems **shall take measures to support the development of AI literacy** of their staff and other persons dealing with the operation and use of AI systems on their behalf (…)"
(EU AI Act, Article 4)

"AI literacy' means **skills, knowledge and understanding** that allow providers, deployers and affected persons, taking into account their respective rights and obligations in the context of this Regulation, to make an informed deployment of AI systems, as well as to gain awareness about the opportunities and risks of AI and possible harm it can cause" (EU AI Act, Article 3)

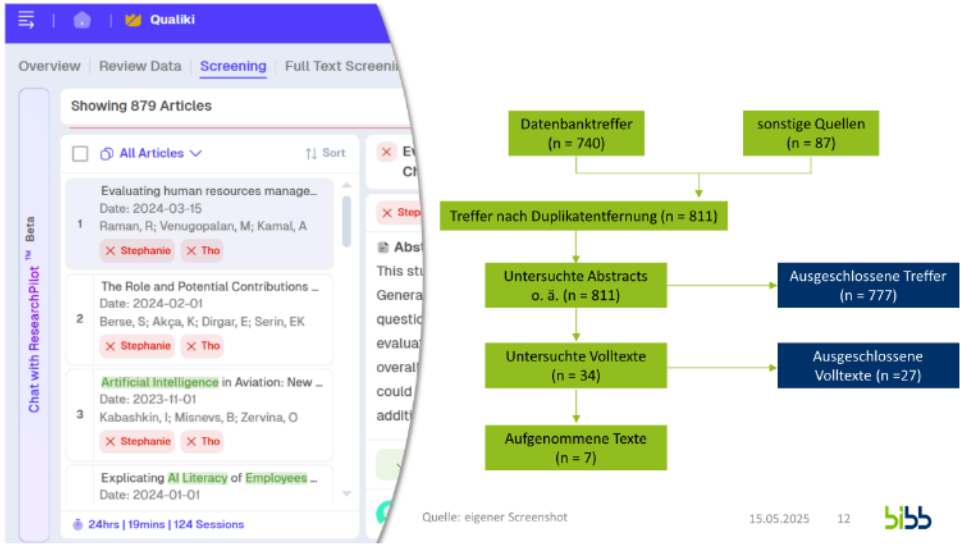
1. **What knowledge and skills do professionals need to use AI systems competently and understand the opportunities, risks, and potential harm associated with AI?**
2. **How does the use of AI in the workplace change competency requirements?**

?

Challenges: Competences



- **What knowledge and skills do professionals need to use AI systems competently and understand the opportunities, risks, and potential harm associated with AI?**
- **How does the use of AI in the workplace change competency requirements?**



Quelle: eigener Screenshot

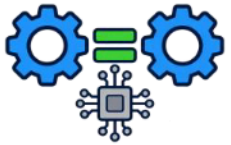
Results of the Scoping Review:

- only a few studies so far
- limited changes in overall skill requirements
- The way AI is used is crucial.
- AI-specific competences and existing domain-specific competences must be considered together
- The impact of AI use on competence and competence-requirements is complex and divers.

Conein, S., & Blum, T. (2025). Veränderung von Qualifikationsanforderungen durch den Einsatz von KI im beruflichen Kontext – QualiKi (Entwicklungsprojekt 2.2.433: Abschlussbericht). Bundesinstitut für Berufsbildung.

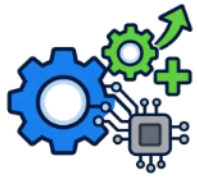
Challenges: Competences

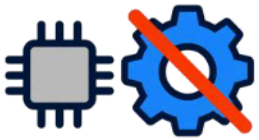
The impact of AI use on competence and competence-requirements is complex and divers



Competence requirements may remain unchanged despite the introduction of AI.

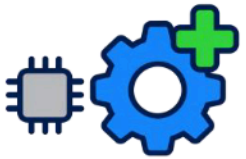
Competence requirements may shift to a higher level.

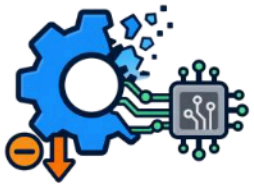




The need for certain competencies may disappear entirely.

Working with AI may itself lead to competence gains among skilled workers.

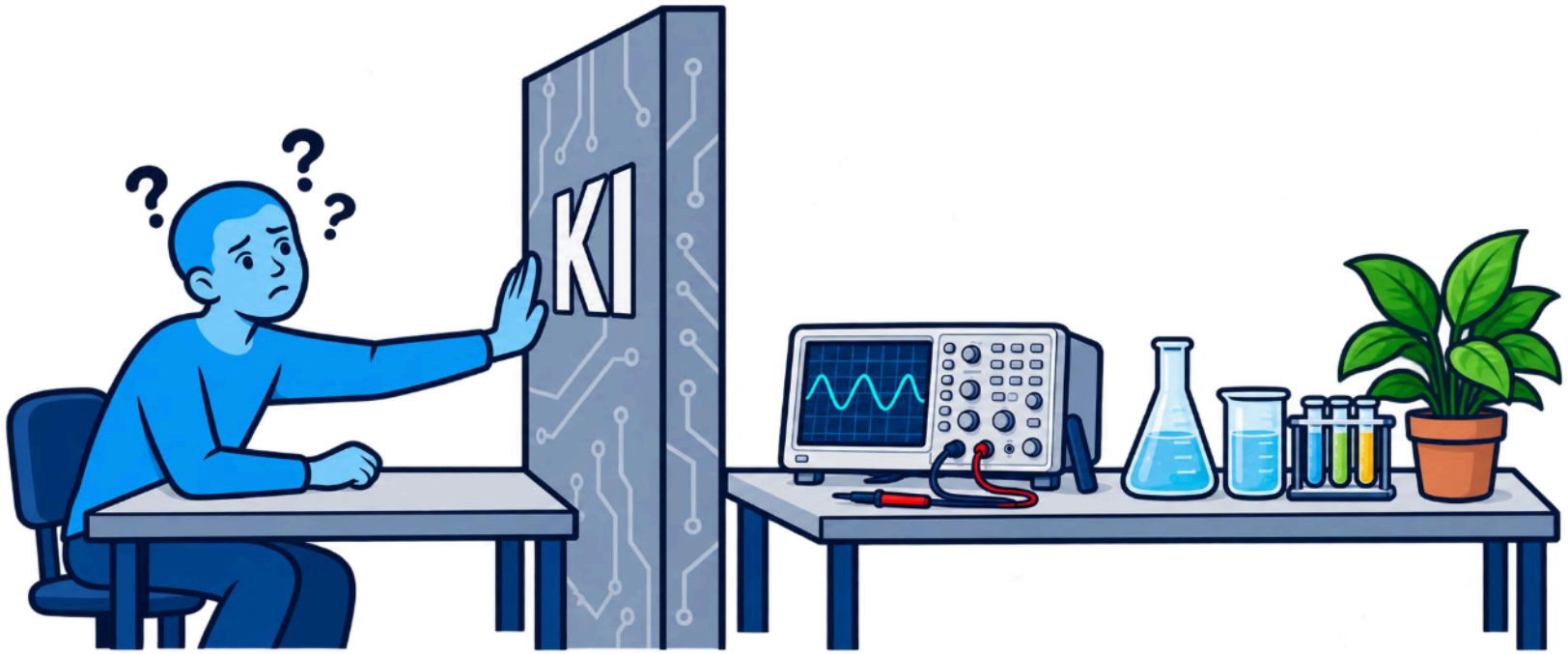


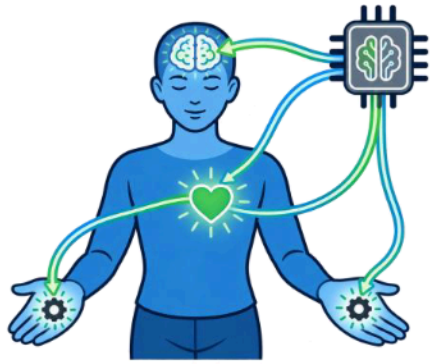


Working with AI may lead to a loss of competence among skilled workers and prevent the development of competence of the apprentices.

Challenges: Competences

Working with AI may lead to a loss of competence among professionals and prevent the development of competence of the apprentices





AI extends into all areas of competence

Working with AI can prevent the development and maintenance of experiential knowledge and, consequently, expertise



Chances & Challenges – don’ts and dos of VET

Don’ts

- Frantically chase an ever-changing list of newly required skills!
- Consider AI as just another digital tool, comparable to others!
- Hastily dismiss previously taught skills as “outdated”. However, be honest about which skills are genuinely no longer relevant!
- Assume there is a one-size-fits-all solution for AI!

Dos

- Make use of AI’s exciting new teaching and learning opportunities and train vocational education and
- training staff accordingly!
- Rethink assessment formats promptly!
- Stay close to workplaces when thinking/investigating the effects of AI!
- Create as many opportunities as possible to build experiential knowledge through real-world experience!
- Advocate for maintaining competence in everyday work, for example via refresher training!
- Keep in mind that AI can potentially affect all competences!

Thank you for your attention!

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All illustrations in this presentation were generated using ChatGPT/OpenAI except for those, for which a source is annotated below the image.

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